Wildlife Information System (WIS): Analysis and Design

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4. Design

4.1 Class Diagram

The class diagram shows the core classes with their attributes and methods. Animals is the aggregate root; Movement_Record, DNA_Sample and Health_Record each belong to one animal, and Users create them.

↗ class diagram · 7 classes

The tables below list every attribute and method; the figure shows the key ones.

Class	Attributes	Methods
Species	speciesId, commonName, scientificName, classification, conservationStatus, description	addSpecies(), updateSpecies(), deleteSpecies(), getSpeciesById(id)
Animals	animalId, speciesId, locationId, name, dateOfBirth, gender, ageYears, weightKg, uniqueMarking, status, addedBy, createdAt, updatedAt	addAnimal(), updateAnimal(), deleteAnimal(), getAnimalById(id), getAnimalsBySpecies(speciesId), getMovementHistory(), getHealthRecords(), getDNASamples()
Locations	locationId, name, latitude, longitude, altitude, region, country, habitatType, description	addLocation(), updateLocation(), deleteLocation(), getLocationById(id)
Movement_Record	movementId, animalId, locationId, recordedAt, latitude, longitude, altitude, speedKmh, source, createdAt	addMovement(), getMovementByAnimal(animalId), getMovementByDateRange(animalId, start, end), deleteMovement(id), bulkInsert()
DNA_Sample	dnaSampleId, animalId, sampleId, collectionDate, sampleType, labName, labResult, speciesConfirmed, notes	addDnaSample(), getDnaSamplesByAnimal(animalId), updateDnaSample(), deleteDnaSample(id)
Health_Record	healthRecordId, animalId, recordDate, recordType, description,	addAnimalRecord(), getHealthByAnimal(animalId),

Class	Attributes	Methods
	veterinarianName, diagnosis, treatment, nextFollowupDate, createdAt	updateHealthRecord(), deleteHealthRecord(id)
Users	userId, fullName, email, passwordHash, role, organization, status, createdAt	register(), login(email, password), logout(), updateProfile(), changePassword()

4.2 Sequence Diagram

The sequence below shows how an external application queries animal data through the API: GET /api/animals?species=Snow+Leopard with an Authorization: Bearer token.

↗ sequence diagram · 6 participants, 9 messages

1. The external app sends the GET request with its JWT in the Authorization header.
2. The API Gateway asks the Authentication Service to validate the token.
3. The Authentication Service confirms the token is valid and returns the user role.
4. The API Gateway forwards the request with its parameters to the Query Controller.
5. The Query Controller passes the parameters to the Animal Service, which constructs the SQL query joining the animals and species tables.
6. The Animal Service queries the database for the relevant rows.
7. The database returns the query result.
8. The Animal Service returns the result as JSON to the Query Controller.
9. The API Gateway responds to the external app with HTTP 200 and the JSON payload.

If step 3 fails, the gateway stops the request and responds with HTTP 401.

4.3 Architecture

WIS follows a 3-tier architecture. External users and applications reach the Application Layer through the RESTful API; the Application Layer handles authentication, business logic and data formatting; the Data Layer stores everything persistently.

↗ 3-tier architecture · 3 layers

Layer	Component	Responsibility
Presentation	HTML/CSS/JavaScript web interface	Data entry, search and export
Application	Node.js + Express.js API server, JWT authentication	Business logic, authentication, query processing
Data	PostgreSQL	Wildlife records, users and audit logs

Key API endpoints

Method	Endpoint	Description
GET	/api/animals	List animals (with filters)
GET	/api/animals/:id	Get one animal by ID
POST	/api/animals	Add an animal record
PUT	/api/animals/:id	Update an animal record
DELETE	/api/animals/:id	Delete an animal record
GET	/api/animals/:id/movements	Movement history for an animal
POST	/api/movements/upload	Bulk upload CSV movement data
GET	/api/animals/:id/health	Health records for an animal
GET	/api/animals/:id/dna	DNA records for an animal
POST	/api/auth/login	Log in and receive a JWT
GET	/api/species	List all species

4.4 Interface Design

WIS has an easy-to-use web interface with a sidebar for Dashboard, Animals, Movements, Health, DNA Samples, Users, Reports and Logout.

Page	Contents
Login	Email field, password field, Remember me, Forgot password link, Login button; shows a red error message for incorrect credentials
Dashboard	Welcome message, statistic cards (total animals, species, movements, health records), a recent-records table and a View All button
Animal List	Searchable, sortable table of all animals with species and location filters, an Add Animal button, paging, and View, Edit and Delete buttons per row
Add Animal Form	Species, name, gender, age (years), weight (kg) and location fields, with Submit and Cancel buttons
Movement Records	Sortable table of GPS coordinates and timestamps for one animal, with an Upload CSV button
API Documentation	Base URL, endpoint list, required parameters and example responses

Page	Contents
User Management (admin only)	Sortable user table (ID, name, email, role, status, created date) with Create User, Edit and Delete; Create and Edit forms with role and status selectors

Design principles: minimal colour, only blue and white; all buttons have descriptive names; errors are shown in red and success messages in green.